CHAPTER 1

# Numerical sequences and discrete models

**Chapter path**

**Objectives.** Move from a word problem to a sequence; distinguish increase and rate; calculate terms and sums; justify a variation; experiment with a threshold.
**Prerequisites.** Algebraic manipulation, integer powers, percentages, affine functions, Python loops.
**Yearly reference point.** 8·17 September: 8 hours, then review

## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Calculate $3 + 2 \times 4$.

2. Calculate $2^0$, $2^3$ and $2^{-2}$.

3. A count of 80 increases by 5%. Give the new count.

4. For $f(x) = 3x - 2$, calculate $f(0)$ and $f(4)$.

5. How many integers are there from 3 to 10 inclusive?

6. A loop `for k in range(4)` takes which values of $k$ ?

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a question similar, then explain the rule used.

**Oral work and logic.** In 90 seconds, explain the difference between index, term and number of terms. Give a counterexample to a rule guessed from three values.

## Investigate before formalising

In pairs, compare two collection schemes: 20 kg in the first week, then 5 kg more each week; 20 kg then a 20% increase each week. Conjecture which collects the most in total in six weeks, then justify. Allow 10 min of research and 5 min of comparison.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

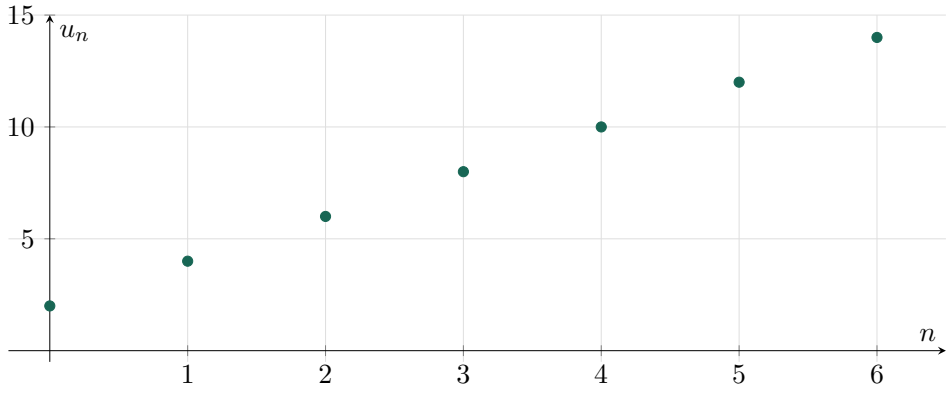
## 1.1 Defining and representing a sequence

**Definition and results**

A real sequence is a function whose variable is an integer: at each index $n$ allowed corresponds a real number $u_n$. The notation $(u_n)$ denotes the sequence, $u_n$ a term. An explicit formula gives directly $u_n$ as a function of $n$. A recursive definition gives a first term and a rule of correspondence. The representation is a set of points $(n; u_n)$ : do not join necessarily these points.

**Status and justification.** Definition. The uniqueness of the terms obtained from a first term and of a deterministic rule is admitted; no formal proof by induction is required.

**Worked example.** For $u_n = n^2 + 1$, $u_0 = 1$, $u_3 = 10$. For $v_0 = 1$ and $v_{n+1} = 2v_n + 1$, $v_1 = 3$, $v_2 = 7$, $v_3 = 15$. We cannot replace $v_3$ by $2 \times 3 + 1$ : the rule refers to the previous term.




Terms of $u_n = 2n + 2$ : only integer indices are plotted.

**Watch out**

The index is not the value. A finite list of terms does not determine a unique rule.

**Immediate application.** Calculate the first four terms of $w_0 = 2$, $w_{n+1} = w_n + 2n + 1$ ; propose an explicit formula.

## 1.2 Arithmetic Sequences

**Definition and results**

A sequence is arithmetic with common difference $r$ if $u_{n+1} = u_n + r$ for every allowed index. Then $u_n = u_p + (n-p)r$ for $n \geq p$. If $r > 0$, it is strictly increasing; if $r < 0$, strictly decreasing; if $r = 0$, constant. The points lie on a line.

**Status and justification.** For $p$ fixed, the sequence $v_n = u_p + (n-p)r$ satisfies $v_p = u_p$ and $v_{n+1} = v_n + r$. By uniqueness of the successive terms, $u_n = v_n$. The variation follows from the sign of $u_{n+1} - u_n = r$.

**Worked example.** A reservoir contains 140 L and receives 12 L per minute. After $n$ minutes, $u_n = 140 + 12n$. At 5 min: 200 L. If we start indexing at 1 with 140 L, we write $u_n = 140 + 12(n-1)$.

**Watch out**

Adding a constant percentage does not in general give an arithmetic sequence.

**Immediate application.** We know $u_3 = 17$ and $u_8 = 37$ for an arithmetic sequence. Determine $r$ and $u_0$.

## 1.3 Geometric sequences

**Definition and results**

A sequence is geometric with common ratio $q$ if $u_{n+1} = qu_n$. For $n \geq p$, $u_n = u_p q^{n-p}$, with the convention that an empty product equals 1. A constant rate $t$ gives $q = 1 + t$. The variation decides with $u_{n+1} - u_n = (q-1)u_n$. For $q < 0$, the signs may alternate.

**Status and justification.** The formula gives the first term and its multiplication by $q$ gives the next term. The uniqueness of the generation justifies the formula. For $q = 0$, all terms after first are zero; deal with this case directly.

**Worked example.** A quantity of 160 mg loses 25% at each step: $u_n = 160(\frac{3}{4})^n$. The terms are positive and $u_{n+1} - u_n = -\frac{u_n}{4} < 0$. We get $u_2 = 90$ mg, without confusing a decrease of 25% and 25 mg withdrawal.

**Watch out**

$q > 1$ is not enough to conclude that every geometric sequence is increasing: the sign of the first term matters.

**Immediate application.** Study the variation of $u_n = -2 \times 3^n$, then of $v_n = 2(-1)^n$.

## 1.4 Sums, indices and proofs

**Definition and results**

For $n \geq 1$, $1+2+\cdots+n = \dfrac{n(n+1)}{2}$. For $q \neq 1$ and $n \geq 0$, $1+q+\cdots+q^n = \dfrac{1-q^{n+1}}{1-q}$. For $q = 1$, the sum is $n+1$. Thus, from $u_p$ to $u_n$, an arithmetic sum equals $\dfrac{(n-p+1)(u_p+u_n)}{2}$; a geometric sum is $\dfrac{u_p(1-q^{n-p+1})}{1-q}$ if $q \neq 1$.

**Status and justification.** Writing $S = 1 + \cdots + n$ in both directions, each pair of the sum 2 $S$ is $n+1$, with $n$ even. For $T = 1 + q + \cdots + q^n$, subtract $qT = q + \cdots + q^{n+1}$ : $(1-q)T = 1 - q^{n+1}$. The case $q = 1$ must come before the division.

**Worked example.** For $u_n = 3 + 2n$, $u_2 + \cdots + u_6 = \dfrac{5(7+15)}{2}$ = 55. For $v_n = 2 \times 3^n$, $v_0 + \cdots + v_3 = \dfrac{2(1-3^4)}{1-3} = 80$.

**Watch out**

Count the terms before using the formula: from 0 to n, there are n+1.

**Immediate application.** Calculate $\sum_{k=2}^{5} 2^k$ without a summation symbol in the answer.

## 1.5 Conjecture a limit and program a threshold

**Definition and results**

A limit is approached intuitively here. The terms $\dfrac{1}{n+1}$ approach 0; $n^2$ grows unbounded; $(-1)^n$ does not settle down. A numerical experiment suggests a conclusion but does not prove a behaviour for all indices. A threshold algorithm must specify the direction of the inequality and plan a stopping condition.

**Status and justification.** Illustrative observations, without a formal theory of limits. The solution of a integer threshold is checked by testing the index found and the previous index, when monotonicity is established.

**Worked example.** For $u_n = 80(\frac{1}{2})^n$, the first index such that $u_n < 6$ is 4: $u_3 = 10 \geq 6$ and $u_4 = 5 < 6$. The programme begins with $n = 0, u = 80$ and repeats the division by 2 while $u \geq 6$.

**Watch out**

An unbounded loop may never stop if the threshold cannot be reached; $<$ and $\leq$ are not interchangeable.

**Immediate application.** For $u_n = 48(\frac{1}{2})^n$, determine the first indices where $u_n \leq 6$ and where $u_n < 6$.

## Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher chooses one. Exercise 8 is revisited later.

### Exercise 1.1 · Getting to grips with indices

Consider the sequence $(u_n)$ defined for every natural number $n$ by $u_n = 2n^2 - 3n$. The first term is therefore the one with index 0.

1. Calculate separately $u_0$, $u_1$ and $u_4$, by showing the substitution of $n$.

2. Express $u_{n+1}$ as a function of $n$, then expand and simplify this expression.

3. Calculate $u_{n+1} - u_n$. By studying its sign for integer indices, describe the variation of the sequence.

### Exercise 1.2 · Choosing the right model

A nursery currently has 100 young plants. It is considering two models to forecast its count: in model A, the number increases by 15 plants per month; in model B, it increases of 10 per cent per month. Let $a_n$ and $b_n$ the projected numbers after $n$ months, with $a_0 = b_0 = 100$. The two models are studied separately.

1. **Model A.** Write the relationship between $a_{n+1}$ and $a_n$. Identify the type of sequence, then express $a_n$ as a function of $n$. Calculate $a_2$.

2. **Model B, after the study of geometric sequences.** Answer the same questions for $(b_n)$ and calculate $b_2$.

3. For each model, explain what is assumed constant. Give one reason why a A very long-term forecast could become unrealistic.

### Exercise 1.3 · Recovering a sequence

The sequence $(u_n)$ is geometric, defined from index 0. Its common ratio $q$ is strictly positive. We knows only $u_1 = 6$ and $u_3 = 54$.

1. Determine $q$ by linking the two known terms. Show all the steps that make it possible to choose its value.

2. Deduce $u_0$, then $u_5$.

3. What would removing the hypothesis change $q > 0$ ? Give all the possible common ratios and explain whether the value $u_5$ would be changed.

## Exercise 1.4 · Rows of seats

A concert hall has 12 rows. The first row has 18 seats; each row next one has two more seats than the previous one. We denote $u_n$ the number of seats in the row $n$, for 1 $\leq n \leq 12$.

1. Define the sequence $(u_n)$ and calculate the number of seats in the last row.

2. Calculate the total capacity of the hall. Explain the number of terms used in the sum.

3. Calculate the mean number of seats per row and use this result to check the total found.

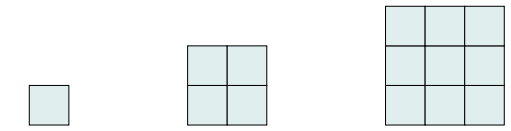
## Exercise 1.5 · Adding up squares

Patterns are built with identical square tiles. Pattern number 1 contains one tile. To go from pattern number $n$ to that of index $n$ + 1, we add 2 $n$ + 1 squares. Let $u_n$ the number of tiles in the pattern at index $n$, for $n \geq 1$.

1. Calculate $u_2$ and $u_3$. Draw an arrangement of these tiles that suggests a formula for $u_n$.

2. Conjecture an expression of $u_n$ as a function of $n$.

3. Check that this expression gives the first term and satisfies the construction rule. Explain why these two checks identify the sequence.



Index 1 Index 2 Index 3

A possible layout for the first three patterns; the drawing helps to formulate a conjecture, which remains to be justified.

## Exercise 1.6 · Two repayment plans

A person must repay 600 monetary units, interest-free. Two plans are being considered. With plan A, she pays 50 units at the end of each month until the loan is repaid. With plan B, she pays each month half of what she still owes. The calculations for model B are done without rounded. We denote $a_n$ and $b_n$ the debts remaining after the payment for month $n$, with $a_0 = b_0 = 600$.

1. Express $a_n$ until the debt is paid off. In which month is the repayment completed?

2. Express $b_n$. Determine the first month in which the remaining debt is strictly less than 10 units ; check the previous month.

3. In model B without rounding, does the debt become exactly zero after a finite number of months? Justify and comment on the difference from a real situation.

## Exercise 1.7 · Reading a list

A program must store the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It starts with the instructions `u=2` and `L=[u]`. It then executes, five times and in this order, `u=3*u-1` then `L.append(u)`.

1. Run the first two iterations of the loop by hand. Then give the contents of the list.
2. How many elements does the list contain after the five iterations? What is the index of the last term recorded?
3. Write the complete program in Python, then check the value of the last element. Explain why the number of terms is one more than the number of iterations.

### Exercise 1.8 · Delayed review

This question is done one week after the chapter, without looking at the formulas. A sequence is defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate $u_7$ stating how many loop iterations separate indices 2 and 7.
2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms of this sum and the number of passes calculated in the first question.

**Independent work.** Do only the task assigned at the end of the session, in the scheme of work Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not use up the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the personal work planned that day (10 to 20 minutes in total).

**CQFD, exercise 9 p. 64, questions a and b.** Calculate the first four terms. Write the substitutions before simplifying.

**CQFD, exercise 11 p. 64, all questions.** Calculate the terms successively and keep the record of the four steps.

## MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $u_n = 3n + 2$ : $u_4$ value

   A. $14$
   B. $18$
   C. $12$
   D. $6$

2. $u_0 = 2$, $u_{n+1} = u_n + 5$ : $u_3$ value

   A. $10$
   B. $12$
   C. $17$
   D. $15$

3. An increase of 8% gives the common ratio

   A. $0{,}08$
   B. $1{,}8$
   C. $0{,}92$
   D. $1{,}08$

4. From $u_3$ to $u_9$ included, we count

   A. $6$
   B. $7$
   C. $9$
   D. $12$

5. $1 + 2 + \cdots + 20$ is

   A. $200$
   B. $420$
   C. $210$
   D. $190$

## MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $u_n = -3 \times 2^n$ is

   A. strictly increasing
   B. constant
   C. Neither increasing nor decreasing
   D. Strictly decreasing

2. 1 + 2 + 4 + 8 equals

   A. $15$
   B. $16$
   C. $14$

D. $8$

3. For $u_n = \dfrac{16}{2^n}$, first index where $u_n < 2$ :

A. $2$

B. $3$

C. $4$

D. $5$

4. The sequence $(-1)^n$

A. tends to 0

B. has no limit

C. is increasing

D. is constant

5. A list contains $u_0$ to $u_{10}$ ; its length is

A. $10$

B. $9$

C. $12$

D. $11$

## True or false? Justify.

1. Every geometric sequence with common ratio $q > 1$ is increasing.

2. If two sequences have equal first three terms, they are equal.

3. For $q = 1$, $1 + q + \cdots + q^n = n + 1$.

**Personal review**

I distinguish explicit and recursive; I recognise a constant increment or rate; I count the terms; I check the index before a threshold.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.